ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-13 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

2. I can use input-output rules to represent relationships.

A machine adds 4 and then doubles the result. The output is 18. Find the input.

3. I can use input-output rules to represent relationships.

Evaluate.

$f(x) = 2x + 5;\ f(-3)$

4. I can use input-output rules to represent relationships.

If $y = 2x - 1$, list outputs for $x = 0, 1, 2$. Then solve $2n + 3 = 11$.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Pairs $(1, 5), (2, 5), (3, 5)$ all have output 5. Is this a function? Explain.

6. I can distinguish functions from relationships that are not functions.

The pairs are $(1, 4), (2, 6), (1, 7)$. Is this relation a function? Explain.

7. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

8. I can distinguish functions from relationships that are not functions.

Is $(1, 3), (2, 3), (3, 3)$ a function? Evaluate $2(-3) + 5$.

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

Find the domain of $\{(-3, 1), (0, 4), (2, 7)\}$.

10. I can identify the domain and range of a relation.

The function $f(x) = \sqrt{x}$ has real outputs. Explain why negative inputs are excluded.

11. I can identify the domain and range of a relation.

x	-2	0	2	4
y	5	1	5	9

List the domain and range without repetitions.

12. I can identify the domain and range of a relation.

Find the range of $\{(1, 4), (2, 4), (3, -1)\}$.

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

14. I can connect linear growth in tables, graphs, and equations.

x	0	1	2	3
A	1	4	7	10
B	1	4	9	16

Which is linear? Give numerical evidence.

15. I can connect linear growth in tables, graphs, and equations.

Complete the table.

x	0	1	2	3
y	4	7	10	?

16. I can connect linear growth in tables, graphs, and equations.

A table has $x = 0, 1, 2, 3$ and $y = 3, 7, 11, 15$. Write a rule.

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

A tank has 12 L at minute 1 and 27 L at minute 6. Find and interpret its average rate of change.

18. I can calculate and interpret slope as a constant rate of change.

One line has slope -2 and another has slope 3. Compare their directions and rates.

19. I can calculate and interpret slope as a constant rate of change.

Find the rise.

(3,−4) to (7,5)

20. I can calculate and interpret slope as a constant rate of change.

Find the slope through (0,3) and (4,11). Then evaluate $f(2)$ for $f(x) = 3x - 1$.